

Exercise 1 – Requirements, Tests, and Collaboration

Learning objectives: Understand how automated tests can help collaboration between developers

Today, we look at some of the positive side effects of creating automated software tests. We explore their relationship to functional and non-functional requirements with the help of a concrete example.

The structure of this exercise is as follows:

1. We collect our initial thoughts on the effects of testing regarding collaboration and requirements
2. We look at a concrete code example of automated tests
3. We rework and refine our initial thoughts and learnings
4. We revisit some of the core contents of last lecture

The tasks in this exercise are intended to be worked on by groups of 2-3 people.

Part 1 – Effects of automated Tests

Discuss the following questions and write down your answers (bullet points only) on a white board.

1. Recap: What is the difference between functional and non-functional requirements?
2. What is the relationship between automated tests and functional requirements?
3. What is the relationship between automated tests and non-functional requirements?
4. How do automated tests help with collaboration between software engineers and non-technical development roles (such as product owners)?
5. How do automated tests help with collaboration in-between software engineers?

Part 2 – Code Example

Look at the given example code. Discuss the following questions in your group and write down your answers in the empty space below.

1. What functional requirements can you glean from this code?
2. What non-functional requirements can you glean from this code?
3. How could these specific automated tests help you to collaborate with another technical or non-technical stakeholder?
4. Can you see a common structure in the automated tests? If yes: What is it and what is the point of such a structure?

Note any additional open questions about the code that you would like to have addressed/clarified.

Part 3 – Revisiting the initial Questions

Discuss the questions from Part 1 update your answers on the white board.

Optional: Copy your answers onto this exercise sheet to take home.

2. What is the relationship between automated tests and functional requirements?
3. What is the relationship between automated tests and non-functional requirements?
4. How do automated tests help with collaboration between software engineers and non-technical development roles (such as product owners)?
5. How do automated tests help with collaboration in-between software engineers?

Part 4 – Revisiting the Lecture

4.1.) Explain why testing is important using two examples **not** from the lecture. Find two known errors and explain how these could have been prevented using testing.

4.2.) Explain in your own words: How do Quality Assurance (QA) and Testing relate to one another.

4.3.) Collaboration: What tools did you use for collaboration / documentation in the last two semesters? Which features did you perceive as useful for collaborating?
